Fundamentals of Big Data Sequence Analysis

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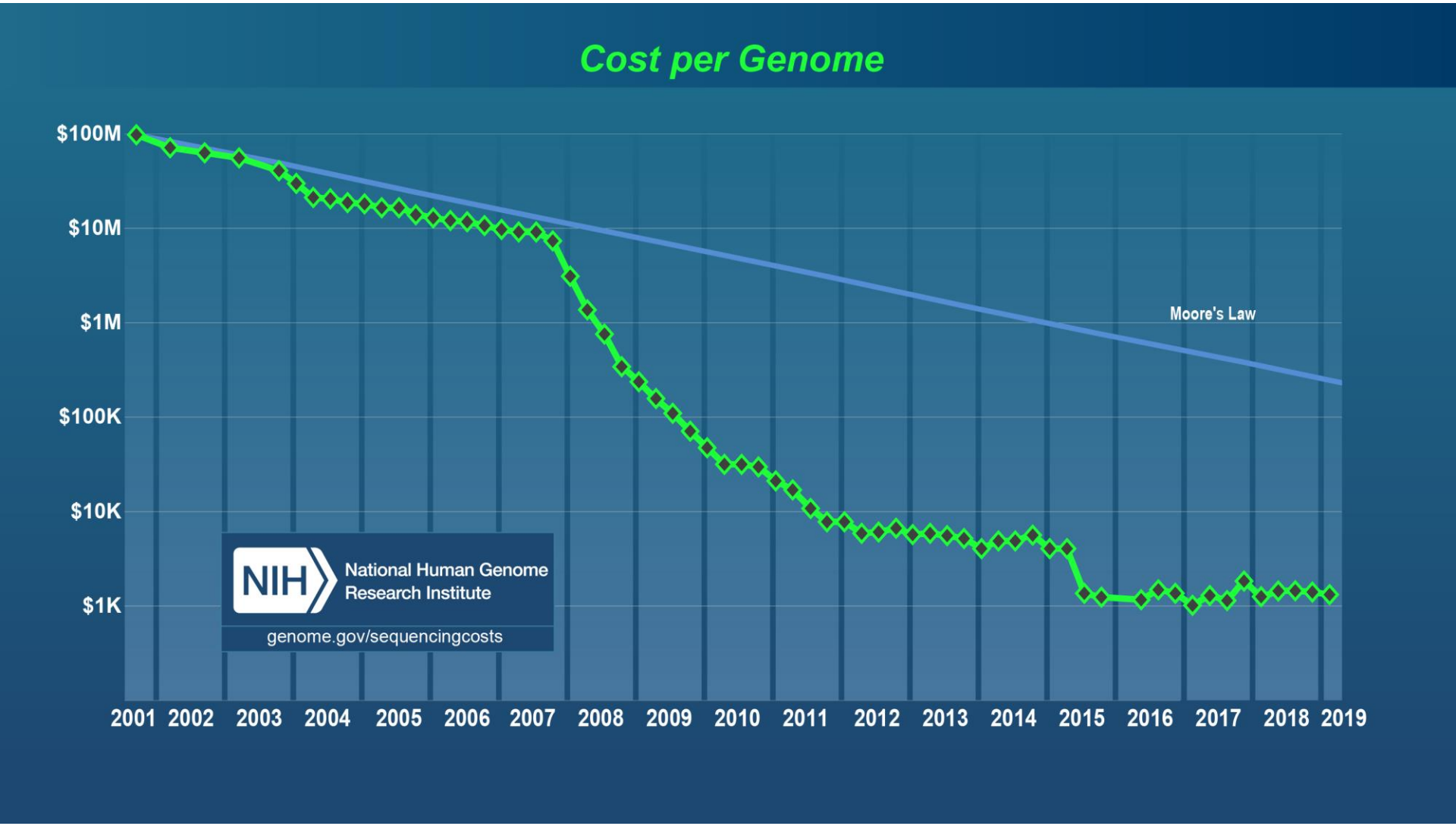
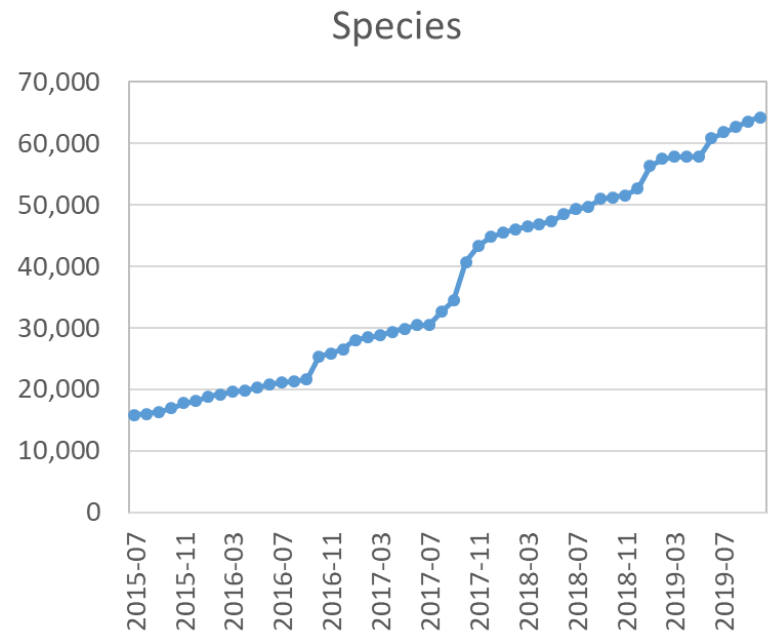
Does the data have to be big?

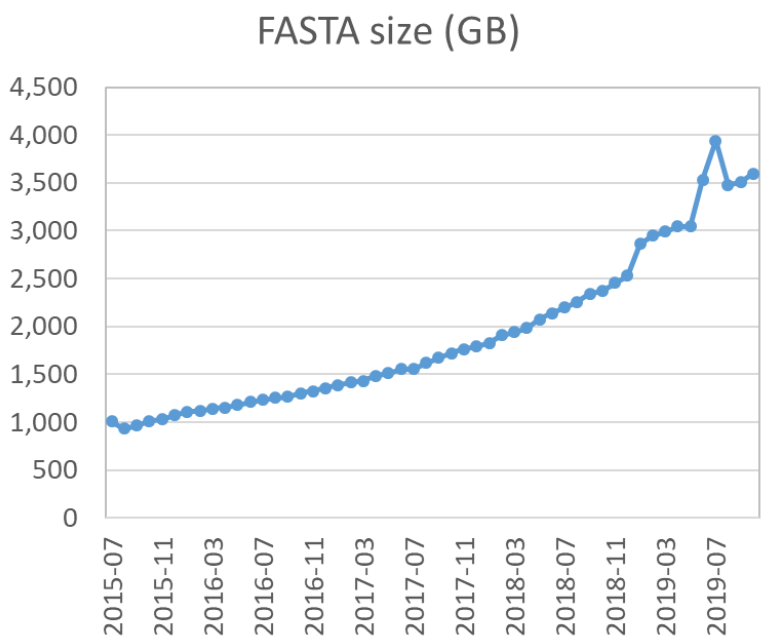


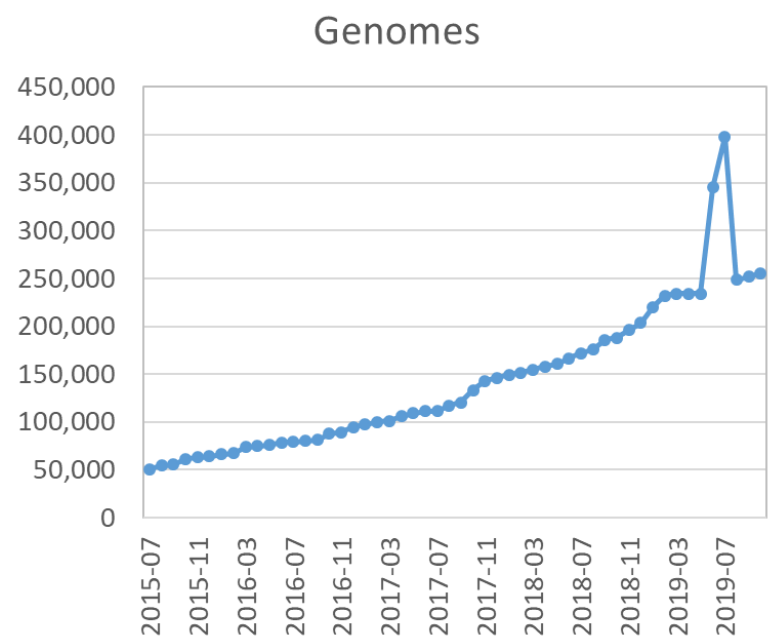
— Yes

Big data

Example:
GenomeSync database
http://genomesync.org/

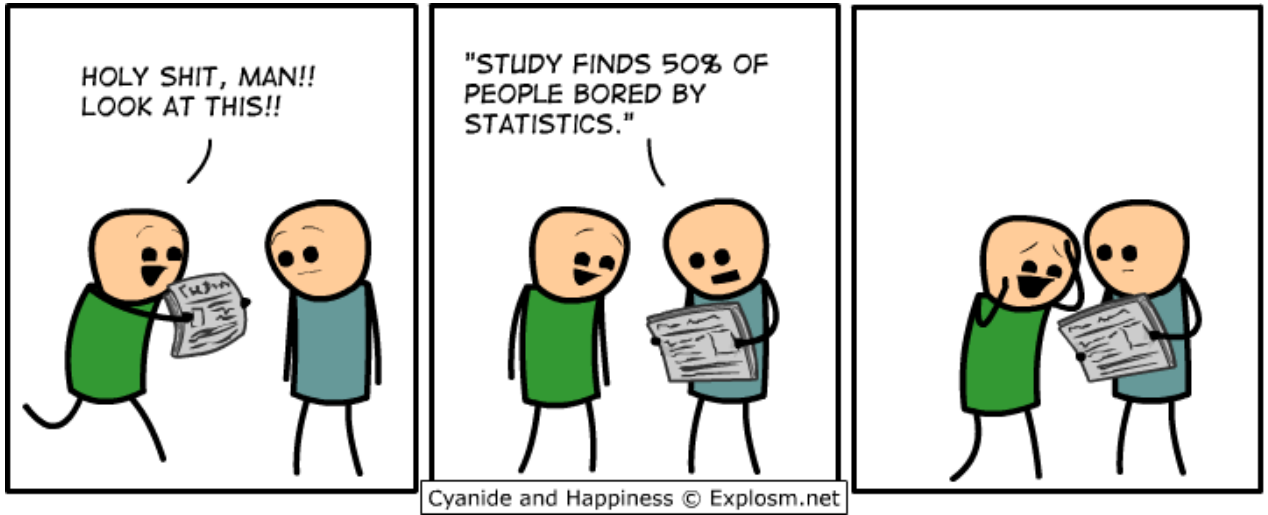






Why work with big data?

1. Sequencing is increasingly affordable
2. Statistical power
3. Interesting questions
4. Useful skills
5. Most important: It's fun.



Principles of big data analysis

Scalability

Scriptability

Reproducibility

(Parallel computing)

In practice

Manual steps		Scipts and programs
Mouse clicks	⇨	Keypresses
GUI		Command line

Data science, expectations:

Data science, reality

edit seq-split-to-lines.c - Far 3.0.5354 x64

C:\prg\seq-tools\seq-tools\src\seq-split-to-lines.c 1252 Ln 8/82 Col 1 Ch 1

```c
#include "common.c"


static unsigned long long line_length = 0ull;


static void done(void)
{
    free_in_buffer();
}


static void process(void)
{
    unsigned long long line_rem = 0ull;

    for (;;)
    {
        in_begin = 0;
        in_end = fread(in_buffer, 1, in_buffer_size, stdin);
        if (in_end == 0) { break; }

        if (line_rem > 0)
        {
            unsigned long long len1 = in_end - in_begin;
            if (len1 > line_rem) { len1 = line_rem; }
            fwrite(in_buffer, 1, len1, stdout);
            fputc(10, stdout);
            in_begin += len1;
            line_rem -= len1;
        }

        for (size_t i = in_begin + line_length; i <= in_end; i += line_length)
        {
            fwrite(in_buffer + in_begin, 1, line_length, stdout);
            fputc(10, stdout);
            in_begin = i;
        }

        if (in_begin < in_end)
        {
            unsigned long long len1 = in_end - in_begin;
            fwrite(in_buffer + in_begin, 1, len1, stdout);
            line_rem = line_length - len1;
        }
    }

    if (line_rem != 0 && line_rem != line_length) { fputc(10, stdout); }
}


int main(int argc, char **argv)
{
    atexit(done);
```

1Help 2Save 3 4Quit 5 6View 7Search 8OEM 9 10Quit 11Plugin 12Screen

Concepts

Terminal (console, shell, command line)

Filesystem, file, directory, relative and absolute path

Commands and arguments, IO redirection

Permissions

Shell scripts

File managers and text editors

Programming languages, compiling and interpreting